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DBbun Invisible Illness Burden Simulator: interoperability status

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Standard or framework	What DBbun does today	What it does not do	Evidence	Next step
HL7 FHIR R4	Every simulator exports its inputs and default results as FHIR R4 Observation resources, with a DocumentReference for the source paper and a Provenance resource. DBbun-defined extensions and code systems ship with the file.	No patient. Not clinical data, not profiled to US Core, not certified. Terminology checking was off.	All 26 files (25 simulators and one reuse demonstration; the second reuse demonstration has no validator record) pass the official HL7 validator (R4 4.0.1) with 0 errors, terminology checking off; 40 to 74 best-practice warnings each. Record: FHIR_VALIDATION.md.	Profile to US Core where it fits; run with terminology checking on.
CMS Interoperability Framework	Echoes its consent and transparency principles at DBbun's own scale: a document's text reaches the model but never the original file; every model call is recorded in the download. The app performs simulator arithmetic in ordinary code; generated Python downloads were run separately during the local reproduction check.	Does not claim formal alignment.	App FAQ ("What happens to my uploaded document?") and the notice file in each bundle.	Review the framework's patient-access requirements when a claims input is built.
Blue Button 2.0	Demonstrated against the CMS sandbox with synthetic test beneficiaries: an OAuth 2.0 sign-in with PKCE, Coverage and ExplanationOfBenefit read as FHIR R4, reduced in memory to yearly totals and shown beside a simulator.	No real Medicare data; sandbox only. No CMS production review. Not endorsed or certified by CMS.	A real run on October 7, 2026 read 155 synthetic claims for 2015 to 2025; a saved example is in the public demo (Evidence tab); README section; tests.	Apply for CMS production access; add consent and data-retention controls for real beneficiaries.
USCDI+	Not implemented.	At the last check (August 2026) USCDI+ had no cost-of-illness domain.	App FAQ.	Tag extracted diagnoses with ICD-10-CM codes so exports map to existing data classes.
TEFCA	Not implemented, by design.	TEFCA exchanges identified clinical records between institutions. DBbun does not implement TEFCA exchange. The local upload workflow rejects documents with Social Security numbers or medical-record-number and date-of-birth labels; this is not a guarantee that every identifier is detected.	App FAQ.	None planned.
DBbun programming interface (REST, OpenAPI)	Submit a paper, collect the result later, call a finished simulator with your own inputs; API keys and a queue.	Not a public self-service upload service; it runs on Dr. Kartoun's computer and answers only there by default.	API.md (public); exercised in the full tool.	Hosted deployment with rate limits.
Reuse for other conditions	The extraction, labeling and checking pipeline runs unchanged apart from a label and a scope sentence. Two demonstrations on other conditions (one without costs, one cost-effectiveness paper) are kept apart from the 25 simulators. The cost paper's 2022 dollar amounts are restated in 2025 prices by the code-side price-index check.	No curated condition-specific federal evidence and no outside validation. The price index is general inflation, not medical-care inflation.	Library section "Reuse demonstrations"; FAQ entry.	Curate federal evidence, comparable definitions and source review for a new condition.

Why it matters for adaptation

Bundles keep each number tied to its source, its label (observed, assumed or projected) and its model call. These links support inspection and reuse by another system; they do not establish that extraction or tagging is correct. That is the reuse case: the pipeline, the evidence labels and the guardrails are not specific to Lyme disease or Long COVID; the curated federal evidence is, and curating it is the work needed for a new condition.